User — Message 1

Inspect this existing Snake game project and its README. Before editing, inspect main.py, game/game_engine.py, game/snake.py, game/food.py, and requirements.txt.

Implement the four README tasks with minimal changes:

1. Prevent direct 180-degree reversal in Snake.set_direction() and ensure collision detection remains correct.

2. Replace the console-only Game Over message with an on-screen Game Over screen showing the final score.

3. Add replay options for Easy, Medium, Hard, and Exit. Reset the snake, food, score, and game state correctly when replaying.

4. Add sound effects when food is eaten and when Game Over occurs. The game must not crash if audio is unavailable.

Preserve the existing project structure, arrow-key/WASD controls, scoring, and food spawning behavior. Do not rewrite the game from scratch. Apply the changes to the actual files, then summarize which files changed and how to test the features. Do not claim tests passed unless you ran them.

GitHub Copilot — Response

Implemented the README tasks in the project at C:/Users/pvnan/45_snake:

- C:/Users/pvnan/45_snake/game/snake.py
  - Prevented direct 180° reversal in `Snake.set_direction()`
  - Kept self-collision checks accurate by ignoring the tail cell when it is moving away
- C:/Users/pvnan/45_snake/game/game_engine.py
  - Added an on-screen Game Over overlay with the final score
  - Added replay menu for Easy, Medium, Hard, and Exit
  - Reset snake, food, score, and movement speed correctly on replay
  - Added safe sound effects for eating food and game over, without crashing when audio is unavailable
- C:/Users/pvnan/45_snake/main.py
  - Initialized Pygame mixer in a way that fails gracefully if sound is unavailable

I did not change C:/Users/pvnan/45_snake/requirements.txt; it was already sufficient.

How to test:

1. Open a terminal in C:/Users/pvnan/45_snake
2. Run:
   - `pip install -r requirements.txt`
   - `python main.py`
3. Verify:
   - Moving right then pressing left does not instantly reverse into the snake’s own body
   - Hitting a wall or self causes a Game Over overlay with the final score
   - Press `1`, `2`, `3`, or `4` (or `E`, `M`, `H`, `X`) to restart at Easy/Medium/Hard or exit
   - Eating food plays a sound, and the game still runs if audio is unavailable

Validation run:

- `python -m compileall .`
- A short smoke test confirmed the direction guard and replay reset logic work (`smoke-ok`)

User — Message 2

The current session is now attached to c:\Users\pvnan\45_snake. Continue the user's original task in this workspace. Do not request another session or workspace conversion.

GitHub Copilot — Response

Done in this workspace: [main.py](C:/Users/pvnan/45_snake/main.py), [game/game_engine.py](C:/Users/pvnan/45_snake/game/game_engine.py), and [game/snake.py](C:/Users/pvnan/45_snake/game/snake.py).

What changed:

- Prevented illegal 180° reversal in `Snake.set_direction()`
  - Pressing the opposite direction no longer flips into the segment behind the snake head
  - Self-collision logic remains accurate while preserving normal movement
- Replaced console-only game-over output with an on-screen Game Over overlay
  - Shows final score
  - Waits for input rather than printing to console
- Added replay selection
  - Easy, Medium, Hard, and Exit
  - Resets the snake, food, score, and speed correctly when replaying
- Added audio feedback
  - Food-eaten sound
  - Game-over sound
  - Gracefully fails if audio is unavailable

Files changed:

- [game/snake.py](C:/Users/pvnan/45_snake/game/snake.py)
- [game/game_engine.py](C:/Users/pvnan/45_snake/game/game_engine.py)
- [main.py](C:/Users/pvnan/45_snake/main.py)

Validation run:

- `python -m compileall .`
- Smoke test for direction guard + replay reset logic
- Result: `smoke-ok`

How to test:

1. In C:/Users/pvnan/45_snake, install deps:
   - `pip install -r requirements.txt`
2. Start the game:
   - `python main.py`
3. Check:
   - pressing the opposite direction while moving does not instantly kill the snake
   - hitting wall/self shows the Game Over screen with the final score
   - `1`, `2`, `3`, or `4` (or `E`, `M`, `H`, `X`) work for replay/exit
   - eating food produces a sound
   - the game does not crash if sound is unavailable

I did not change [requirements.txt](C:/Users/pvnan/45_snake/requirements.txt) because it already met the project requirements.